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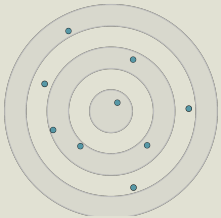
Stabilizing regression coefficients with the lasso

Sometimes linear regression isn't a reliable technique even when the data being modeled has a linear relationship. This happens when the modeling process has high variability, meaning that the coefficients it produces may be very different from sample to sample.

The lasso (**L**east **A**bsolute **S**hrinkage and **S**election **O**perator) is one of the most basic and important ways of addressing this. The idea is to reduce the variability in the model by introducing a small amount of bias.

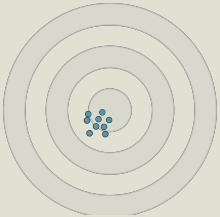


A model generated from sample data is akin to a dart thrown at a dartboard. The sampling process introduces randomness, so darts don't generally hit the bullseye (the true parameter values for the process being modeled).



This variability can substantial. In this picture, the darts hit the bullseye on average, though many of them don't land anywhere close.

The lasso doesn't mind if the model misses a little on average, as long as results don't vary too much from sample to sample.



This is one form of the **bias-variance tradeoff**.

Recall that a linear regression model $y_i \sim \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}$ is fit by minimizing the sum of the squared residuals across all possible values of the coefficients $\beta_0, \ldots, \beta_p$.

$$\mathrm{RSS} = \sum_i (y_i - \hat{y}_i)^2$$

Here y_i is the observed value of the response variable while $\hat{y}_i$ is the value predicted by the model.

Lasso regression minimizes a slightly different quantity.

$$\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^{p} |\beta_j|$$

The first part of this expression is the same residual sum of squares used in simple linear regression. The second part represents a penalty on larger coefficient values (excluding the intercept), which has the effect of shrinking their optimal values somewhat.

Because λ isn't determined directly by the training data, it's referred to as a **hyperparameter**.

$$\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^{p} |\beta_j|$$

An optimal value is typically chosen using cross-validation.

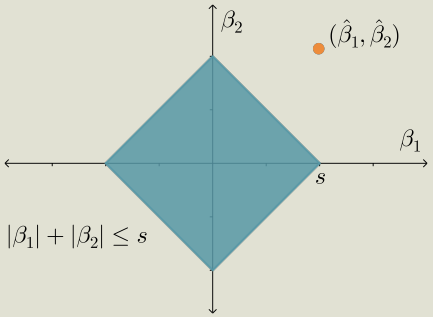
It can be shown that minimizing $\sum(y_i - \hat{y}_i)^2 + \lambda \sum |\beta_j|$ over all possible coefficient values is equivalent to minimizing RSS subject to

$$\sum |\beta_j| \leq s,$$

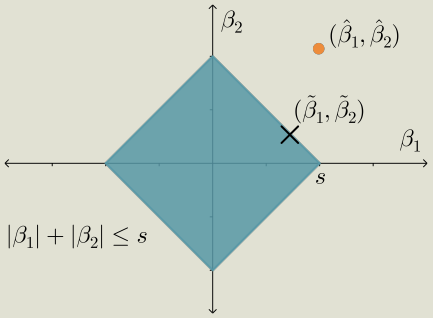
where s is uniquely determined by λ. In other words, lasso regression looks for the best possible values of $\beta_1, \ldots, \beta_p$ within a hypercube centered at zero.

This formulation is sometimes more intuitive because it comes with a picture.

The plot below shows the two-variable case, though it would look similar in any number of dimensions. The shaded teal region represents the set of allowable (β_1, β_2), while the orange point $(\hat{\beta}_1, \hat{\beta}_2)$ represents the coefficients that globally minimize RSS.



The model chooses parameter values $(\tilde{\beta}_1, \tilde{\beta}_2)$ that minimize RSS within the allowable set.



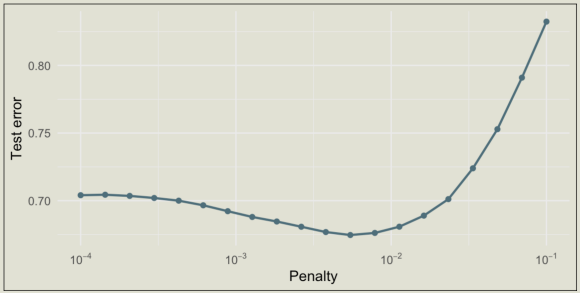
This doesn't always mean the geometrically closest point.

When $\lambda = 0$, there is no penalty and the lasso reduces to ordinary least squares regression. The allowable region is the entire parameter space. As λ increases, that region shrinks.

Typically, test error decreases even after the allowable set no longer includes $(\hat{\beta}_1, \hat{\beta}_2)$ but soon begins to increase again.

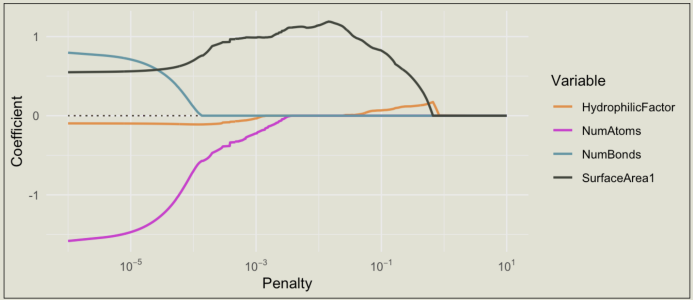
Example. The `AppliedPredictiveModeling` R package includes many data sets to support the excellent book by Kuhn and Johnson, including `solubility`, in which approximately 1000 observations of more than 200 variables can be used to model the solubility of chemical compounds. The package also includes a holdout set of about 300 observations.

As λ increases from zero, the performance of a lasso model improves until the benefit of reduced variability is outweighed by the cost of added bias.



In this case, the optimal value is approximately $\lambda = .005$.

The following plot shows the coefficients of four predictors as λ increases.



At the left, the penalty is near zero, so the model is essential just an OLS regression. At the right, the penalty is so large that all coefficients are zero.

Scaling a variable in linear regression also scales the corresponding coefficient. For instance, converting a variable x_i from dollars to cents would multiply β_i by a factor of $1/100$.

The lasso penalty, $\lambda \sum |\beta_j|$, is harsher toward variables with larger values, which means it's sensitive to the relative scales of the variables.

For this reason, explanatory variables in a lasso model are usually *standardized* prior to fitting. That is, each predictor is converted to a z-score,

$$z_{ij} = \frac{x_{ij} - \overline{x}_j}{s_j}.$$

Here $\overline{x}_j$ and s_j represent the mean and standard deviation of the jth variable while x_{ij} represents the ith observation of that variable.

Untransformed

HydrophilicFactor	NumAtoms	NumBonds	SurfaceArea1
−1.61	3.37	3.43	6.81
−0.44	3.91	3.97	9.75
−0.38	3.53	3.53	8.25
−2.37	3.30	3.30	0.00
−0.07	3.47	3.47	9.91
−0.95	3.50	3.47	6.00
−1.15	3.58	3.66	8.87
−1.51	3.66	3.64	6.88

Transformed

HydrophilicFactor	NumAtoms	NumBonds	SurfaceArea1
−1.11	0.42	0.53	0.02
0.01	1.62	1.63	0.68
0.07	0.77	0.72	0.34
−1.84	0.27	0.24	−1.49
0.37	0.64	0.59	0.71
−0.48	0.71	0.59	−0.16
−0.67	0.90	1.00	0.48
−1.02	1.07	0.94	0.04

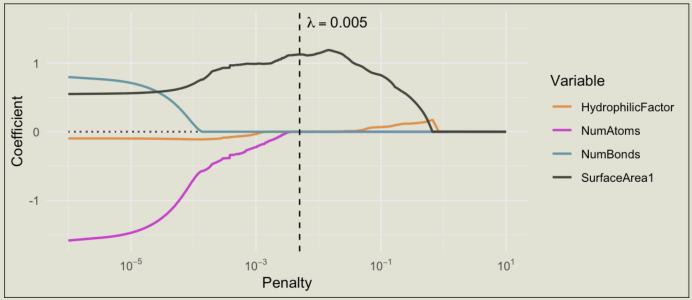
The tables above show the results of standardizing several of the variables in the `solubility` training set.

Once a value of λ has been chosen, maximum likelihood estimation yields coefficients $\tilde{\beta}_0, \ldots, \tilde{\beta}_p$ just as it does with OLS regression. The result is again a linear model,

$$y_i = \tilde{\beta}_0 + \tilde{\beta}_1 z_{i1} + \cdots + \tilde{\beta}_p z_{ip}.$$

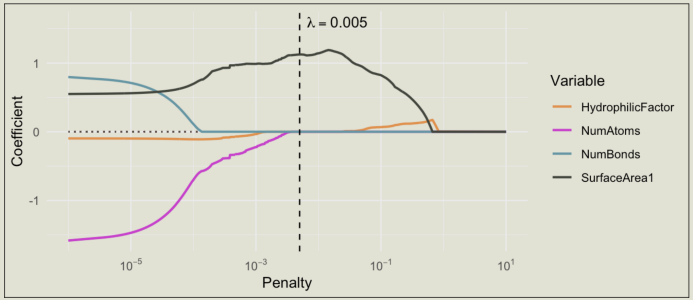
Remember that since standardization happens before model fitting, the coefficients β_j are applied to z-scores, not the original variables in the data set.

The optimal λ in a lasso model often forces some variable coefficients to zero even when they are correlated with the response variable.



This is the case in the `solubility` example.

In this set, `HydrophilicFactor` and `Solubility` have correlation of nearly 40%, yet the finalized lasso model omits `HydrophilicFactor` entirely.



In this sense, the lasso can perform **model selection**, expressing opinions about which variables really have explanatory power and which are merely correlated with ones that do.

In the `solubility` data, for instance, the finalized model includes `SurfaceArea1` but not `HydrophilicFactor`. These variables have a correlation of almost 75%.

When $\lambda = .005$ in the `solubility` model, only 143 of the 228 variables have non-zero coefficients. The other 85 have been removed from the model.

MolWeight	-0.6395399849
NumAtoms	0.0000000000
NumNonHAtoms	0.0000000000
NumBonds	0.0000000000
NumNonHBonds	-0.8497858721
NumMultBonds	-0.2507717866
NumRotBonds	-0.1590003192
NumDblBonds	0.0000000000
NumAromaticBonds	-0.0682813073
NumHydrogen	0.1187408534

This table shows a handful of the finalized coefficient values.

Lasso regression is helpful when a model has high variance, typically because at least one of the following is true:

- The number of predictor variables is large relative to the number of observations.
- The predictors are highly correlated with one another.

Each of these can be checked directly in the training set.

In many machine learning applications, there is little downside to choosing the lasso over unpenalized regression. If a penalty isn't helpful for a particular data set, the tuning process will generally settle on a value of λ very close to zero.

While there is increased computational cost associated with the tuning process, which requires fitting models for a variety of λ values, the added processing time is usually small compared to other machine learning techniques.

Shrinkage methods like the lasso can be applied in other contexts than simple linear regression. For instance, logistic regression models data with a binary response y_i by maximizing the log-likelihood function

$$\ell(\beta) = \sum_{i=1}^{n} \Big[y_i \log p_i + (1 - y_i) \log(1 - p_i) \Big]$$

over all possible coefficient values. To introduce a lasso penalty, we instead maximize

$$\ell(\beta) - \lambda \sum_{j=1}^{p} |\beta_j|.$$

The lasso is just one method of stabilizing regression coefficients through shrinkage. A major alternative is the **ridge regression**, which seeks to minimize

$$\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^{p} \beta_j^2.$$

It's also possible to incorporate both penalties using an **elastic net**.